

brief_revise_agent Factorial Hill-Climb

LLM and structural factor analysis vs. aus_agent_v2

2026-08-07 – 15-topic TREC RAG 2026 dev subset – 21 scored cells – \$340 of \$400 budget

1 Executive summary

brief_revise_agent (a light fork of aus_agent: a pre-flight requirements brief plus one review-and-revise pass on top of the shared agent_harness research loop) was hill-climbed against aus_agent_v2 (this repo’s strongest, most expensive system) via standalone rubric scoring on a fixed 15-topic development set, with sol (gpt-5.6-sol) as design authority and the current session as orchestrator/implementer.

Result: did not beat aus_agent_v2. The best configuration found – gpt-5.6-sol as main generator, round-B adjacent-page fetch on, gpt-5.6-luna as brief-analyst and reviewer, $k = 10$, iteration-1 structure – scores **2.267/3** on standalone rubric grading (the best of every cell tested) but still loses to aus_agent_v2 in a head-to-head arena match, **18–12 (60/40)**, order_consistency 0.733 (4W–7L clean, 4 non-order-consistent). Eleven further single-factor moves against this base – five generic agent_harness toggles, one factor interaction, three brief-analyst model swaps, adjacent-fetch removal, and a novel post-draft “closure critic” mechanism – returned zero improvements (§4.2, §5.2). k and the reviewer model were held fixed throughout and were never themselves varied as a factor. Of the 16 factors in the underlying taxonomy (§3), 8 were empirically scored this session; the base cell is a confirmed local optimum **on the factors actually tested**, not a claim about the full factor space.

2 Method

2.1 Design

Sol-directed hill-climbing: starting from brief_revise_agent’s best known configuration, one factor was changed at a time, scored, and kept only if it improved on the current best – in contrast to the session’s earlier factorial-grid framing, which was superseded once the hill-climb approach was adopted partway through. The full factor taxonomy that this hill-climb draws its factors from is developed in §3; §4 states, per factor, whether it was scored this session, scored in a different session/thread, or never wired in brief_revise_agent at all.

2.2 Evaluation

Primary: standalone rubric grading. One judge call (gpt-5.6-terra) per (cell, topic) grades every official TREC RAG 2026 rubric criterion (~31 criteria across 6 axes, 0–2 each) plus a holistic 0–3 overall score – no opponent answer needed. Chosen over pairwise arena judging because it halves judge calls per cell and produces a genuinely ordinal per-criterion signal instead of a categorical win/loss/tie. Answer text is truncated to 16,000 characters before grading (rubric_scorecard_aus_agent_vs_facets_agent.py); no answer scored this session was observed to hit that cap, but it is not independently verified for every cell.

Confirmatory: arena. One pairwise comparison (gpt-5.6-terra judge, both battle orders) of the best cell against aus_agent_v2, retained as the direct test of the actual competitive objective.

Judge/generator overlap. gpt-5.6-terra is both the sole judge for every score in this report and one of the five generator models compared in §5.1 (its own cell scored 1.867). This is a same-family-model confound that was not controlled for or discussed at the time; it is disclosed here rather than silently assumed away.

Noise floor. overall is a single integer 0–3 graded per topic and averaged over 15 topics, so every reported score is an exact multiple of $\frac{1}{15} \approx 0.0667$ (e.g. $2.267 = 34/15$, $2.200 = 33/15$, $2.133 = 32/15$). The best cell was regenerated once on the same 15 topics; the rerun’s score moved by exactly one topic-grade, i.e. $\frac{1}{15} = 0.067$, purely from generation stochasticity. This single-rerun value ($n = 1$) is used throughout as the noise floor – an effect below it is not distinguishable from one topic’s grade flipping, not a statistically estimated confidence interval. Concretely, “−0.134” means two of fifteen topics scored one point lower; it is not a smaller-than-it-looks effect once read this way.

2.3 Constraint discovered mid-session

The full TREC RAG 2026 research-rubrics development topic set is only 30 topics total. Fifteen were already committed to the tuning set, leaving only 15 unused topics for replication – smaller than originally planned, and insufficient for a full independent held-out arena confirmation once `aus_agent_v2`’s own per-topic generation cost (~\$36.5/topic) is accounted for. The judging budget cap itself was also raised mid-session, from an original \$50 (design) + \$50 (evaluation) split to \$400, once the model-effect result in §5.1 made further hill-climbing look worthwhile (§12 of the narrative worklog).

3 Factor taxonomy

`brief_revise_agent` and its shared `agent_harness` layer expose 16 distinct factors that can plausibly change answer quality: 13 structural/architectural factors (S1–S13) and 3 model-role factors (M1–M3, which model runs which role). This taxonomy was produced by an independent draft-then-review pass (`gpt-5.6-luna` drafted from every system’s README and 16 `aus_agent_v2` module docstrings, `gpt-5.6-terra` reviewed against the same source material and corrected three items) and is reproduced in full at `worklogs/assets/2026-08-07-terra-factor-taxonomy-final.md`. §4 states which of these 16 were actually scored this session.

3.1 Structural factors (S1–S13)

ID	Mech.	What it is	Coverage this session
S1 requirements_brief	PROMPT	Pre-flight tool-less call that extracts explicit/inferred requirements; rendered as an Appendix A block in the main system prompt.	Always ON; never toggled
S2 requirements_brief_schema	SCHEMA	Output contract for the brief: v1 (≤ 8 entries, ≤ 4 implicit, lexical anti-hunch check) vs. v2 (14 entries, ≤ 6 expert-completion entries, no anti-hunch check).	Tested pre-session (v2 regressed vs. v1); v1 is current, not retested here
S3 brief_word_budget	PROMPT	Whether the brief also requests target_total_words (500–1000) and per-requirement target_words; invalid values fail open to no guidance.	Tested in the rounds-B/C/D thread as round C: 0W/13L/2A vs. baseline, worst of all rounds – not retested in this factorial thread
S4 review_revise_pass	CONTROL_FLOW	One-shot pre_final_hook after a valid draft: scans uncited sentences, grades brief requirements, gets reviewer feedback, allows exactly one revision turn.	Always ON (defines brief_revise_agent); its content was widened this session (see closure critic below), never toggled off
S5 adjacent_page_augmentation	RE-TRIEVAL	Fetches the page immediately before/after the top-5 paginated hits of each search batch, zero LLM calls (ported from aus_agent_v2/search.py).	TESTED – off vs. on, sol & qwen
S6 retrieval_engine_set	RE-TRIEVAL	Which retrieval backends (semantic, keyword, hybrid, ssr, lucene_bool) the search tool may use.	TESTED – default semantic, keyword vs. widened +ssr+lucene_bool
S7 search_k	RE-TRIEVAL	Hits requested per search call when the model omits k.	Held fixed at 10; never varied
S8 search_result_filter	RE-TRIEVAL	Post-search relevance pass (minimize_filter/rank_filter) that narrows or annotates returned evidence.	NOT WIRED – both filters key off a per-call requirement field that brief_revise_agent’s search tool schema does not collect; running it as-is would silently degenerate, not test the factor
S9 search_preview_policy	RE-TRIEVAL	Whether staged search text is full, or truncated to a positional preview (no generator model) at a fixed 20,480-char cap.	TESTED – full staging vs. 20,480-char positional cap
S10 stage_search_results	RE-TRIEVAL	Whether ordinary search results are inserted into the staged-context ledger at all (get_documents stays available either way).	TESTED – staged (default) vs. unstaged
S11 judge_relevance_tool	SCHEMA	A 4th advertised tool; when called it opens a separate single-turn conversation asking whether evidence supports a named requirement.	TESTED – off vs. on, alone and in combination with S12
S12 commit_release	SCHEMA	Adds a release array to the commit_context schema so the model can retroactively drop an earlier committed document a new one supersedes.	TESTED – off vs. on, alone and in combination with S11
S13 historical requirement-screener	RE-TRIEVAL	Reverted mechanism: mandatory requirement field on every search call + one-shot DIRECT/LEAD/OFF_TOPIC screener with a retention floor.	Tested pre-session (iteration 3): 13 clean losses vs. 10 for baseline – reverted, not restorable without new code

Table 1: Structural/architectural factor taxonomy. Coverage color: **green** = scored this factorial session, **red** = not wired / not testable as-is, black = tested in a different session/thread or never varied.

3.2 Model-role factors (M1–M3)

Current `brief_revise_agent` constructs the main agent, brief analyst, and reviewer from the **same** backend/model factory call by default; `--model`, `--brief-model/--brief-backend`, and `--review-model/--review-backend` decouple them (added this session to support M2/M3 testing). Five model IDs are confirmed usable: `gpt-5.6-luna`, `gpt-5.6-terra`, `gpt-5.6-sol` (OpenAI backend), `openai.gpt-oss-120b-1:0`, `qwen.qwen3-next-80b-a3b` (Bedrock, `ap-southeast-2` and `us-east-1/us-west-2` respectively).

ID	Role	Coverage this session
M1 <code>main_research_writer_model</code>	Runs the continuous research/writing loop: decomposition, search, context commitment, gap follow-up, final cited report.	TESTED – all 5 models, \$5.1
M2 <code>requirements_brief_analyst_model</code>	Produces the requirements brief (S1) and word-budget guidance (S3) if enabled.	TESTED – terra/gpt-oss-120b/qwen vs. luna, main=sol fixed
M3 <code>reviewer_model</code>	Grades the candidate against the brief and evidence, emits the one bounded revision.	Not tested – explicitly skipped for budget (worklog §14): same role-swap pattern as M2, which showed zero effect

Table 2: Model-role factor taxonomy.

3.3 Mechanism examples

Four representative factors, showing exactly what “the factor” is at the code/prompt level:

S5 adjacent_page_augmentation (RETRIEVAL, zero LLM calls) – `adjacent_pages.py`’s core routine, ported from `aus_agent_v2/search.py`:

```
def adjacent_page_ids(unit_ids: list[str], *,
                     max_seed_hits: int = DEFAULT_MAX_SEED_HITS) -> list[str]:
    """Stable, deduplicated +/-1 page ids for the top `max_seed_hits`
    paginated hits, excluding ids already present in `unit_ids`."""
```

Wired as `search_result_augment` on the shared harness; `--disable-adjacent-pages` turns it off (default: on).

S12 commit_release (SCHEMA) – `commit_release_tool.py` adds exactly one property to the shared `commit_context` schema, deliberately **not** importing `facets_agent`’s bundled variant (which couples release to an unrelated coverage/ready_to_report ledger – a different factor):

```
_RELEASE_PROPERTY: dict[str, Any] = {
    "release": {
        "type": "array",
        "description": (
            "Previously committed ids to drop because a document you are "
            "committing in THIS SAME call supersedes them ..."
        ),
        "items": {"type": "object", "properties": {
            "id": {"type": "string"}, "reason": {"type": "string"}},
            "required": ["id", "reason"]},
    },
}
```

Enabled by `--commit-release`; the harness’s own `apply_commit` already handles a release argument whenever present, so this module only needs to **advertise** the field.

S4 review pass, closure-critic extension (PROMPT, this session’s only new mechanism) – `pre_final_hook` fires at most once by harness design, so this extends the existing single review pass’ issue taxonomy rather than adding a second stage. Base issue types (`review.py`): `ISSUE_TYPES = frozenset({"UNCITED_CLAIM", "WEAK_SENTENCE"})`. With `--closure-critic`, two more are unioned in – `CLOSURE_ISSUE_TYPES =`

frozenset({"UNSUPPORTED_CLAIM", "CONTRADICTION"}) – and the reviewer prompt gains these two bullets (prompts.py, REVIEW_PROMPT_WITH_CLOSURE):

- UNSUPPORTED_CLAIM: a cited sentence that claims MORE than its cited evidence actually states (a number, scope, or causal claim the evidence text does not contain) -- an overclaim, not a missing citation.
- CONTRADICTION: two sentences in the draft that cannot both be true, or a sentence that contradicts its own cited evidence text.

Default closure_check=False is byte-identical to pre-session behavior.

S9 search_preview_policy (RETRIEVAL, no generator model, this session's tested level) – --search-preview-chars 20480 caps each staged search result to 20,480 characters before it enters the ledger; get_documents remains the full-text read route regardless. This is a positional truncation only – the taxonomy's other preview level (an LLM-generated query-relevant snippet, tested inconclusively in facets_agent's Phase 4d) was not exercised here.

4 Results

4.1 All scored cells (§5.1)

Figure 1 ranks every one of the 21 cells scored this session by standalone overall score. The four cells tied at the top (2.267) are the base cell and three factors that made no measurable difference (commit_release, judge_relevance_tool + commit_release together, and the closure critic). Two caveats on comparability across this ranking: the two *-new15 cells (sol/luna replication) are scored on a **different** 15-topic set than the other 19, so their rank position is not a like-for-like factor comparison; and the historical brief-revise-iter1-exp15 reference (2.067) predates round B's commit and therefore lacks S5, conflating a model question with a structural one if read as a pure luna-vs-sol comparison (the apples-to-apples luna comparison is br-luna-current-code-exp15, 1.867, discussed in §6.2).

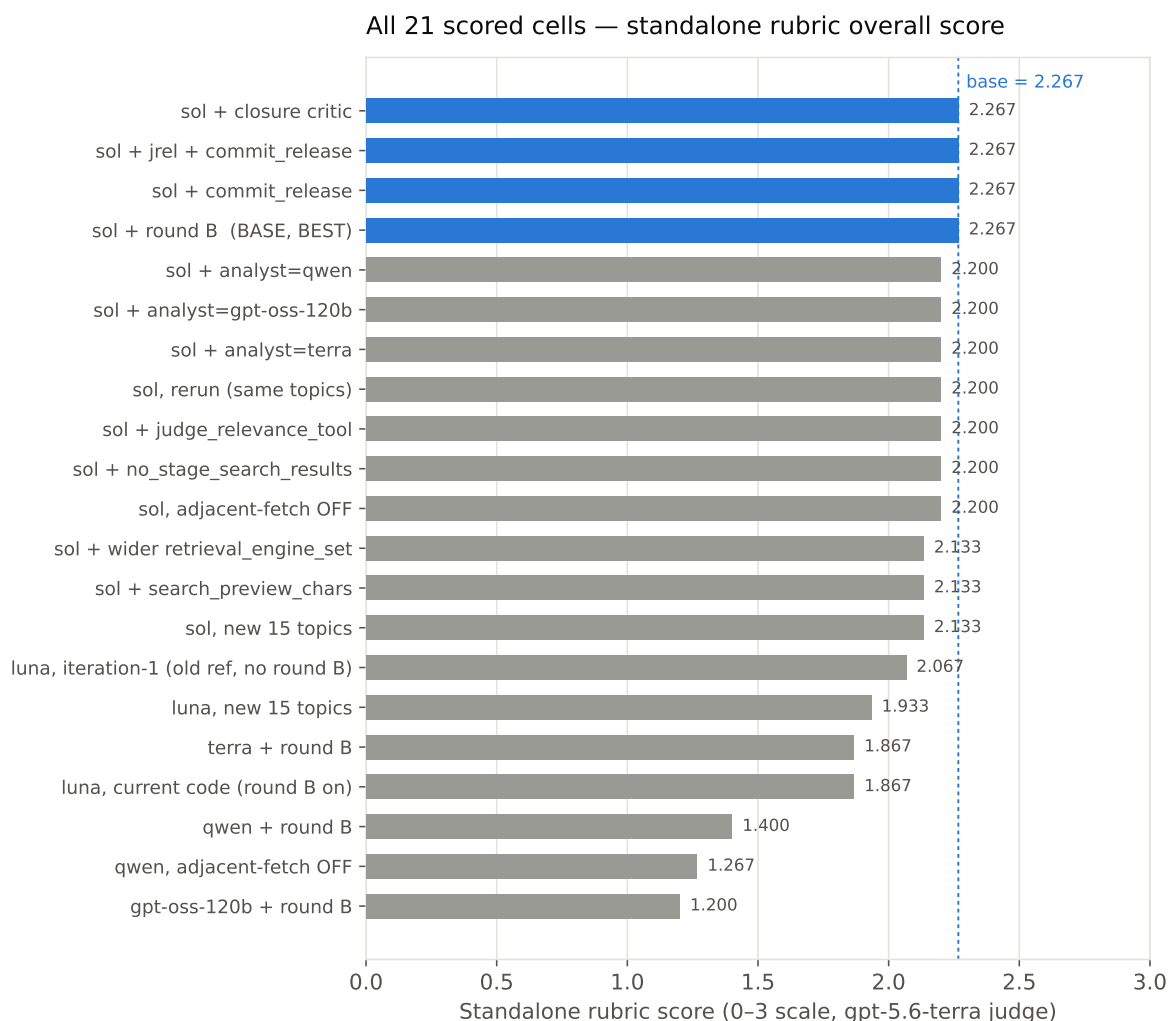


Figure 1: Standalone rubric overall score, all 21 cells, sorted. Base cell in blue. [Interactive version →](#)

4.2 Hill-climb moves and factor effects

Table 3 lists every single-factor move scored against the base cell (2.267), i.e. the moves underlying the “eleven further moves” claim in the executive summary – the narrative worklog’s own running tallies of “6”, “11”, then “12” moves (its §11, §15, §16) reflect moves being added mid-session and one double-count of the interaction cell; this table is the reconciled, deduplicated count of distinct scored comparisons.

Factor	Taxon- omy ID	Δ vs. base	Verdict
Adjacent-fetch OFF (sol)	S5	−0.067	reject
stage_search_results=False	S10	−0.067	reject
judge_relevance_tool=True	S11	−0.067	reject
search_preview_chars=20480	S9	−0.134	reject
wider retrieval_engine_set	S6	−0.134	reject
commit_release=True	S12	0.000	reject (tie)
judge_relevance + commit_release	S11+S12	0.000	reject (tie); no interaction rescue of either factor's own null/negative result
brief-analyst = terra	M2	−0.067	reject
brief-analyst = gpt-oss-120b	M2	−0.067	reject, identical to terra
brief-analyst = qwen	M2	−0.067	reject, identical to the other two
closure critic (S4 + UNSUPPORTED_CLAIM/ CONTRADICTION)	S4	0.000	reject (tie)

Table 3: Eleven distinct hill-climb moves against the base cell (2.267). Noise floor: ± 0.067 (one topic-grade, $n = 1$).

Figure 2 groups the same moves (plus the generator-model comparison that established the base cell in the first place) by factor family. Generator model is the only family whose effects consistently and substantially clear the noise band; every other family in Table 3 clusters at or inside it. One caption note: the “adjacent-fetch OFF, qwen” point is computed against the **qwen** base cell (1.400), not the sol base cell (2.267) plotted elsewhere in the same figure – it isolates S5’s effect for qwen specifically, not a comparison to the session’s overall best cell.

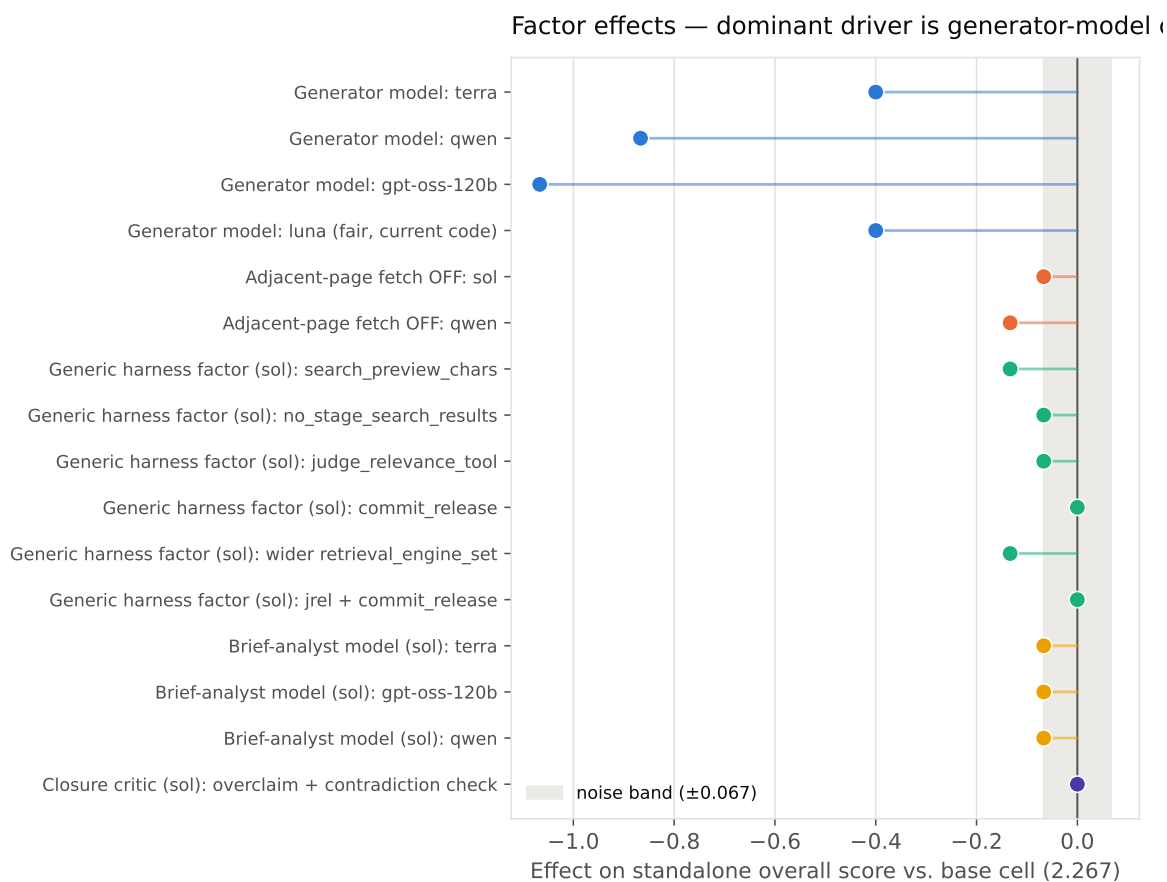


Figure 2: Factor effects vs. base cell (sol, 2.267; qwen points vs. qwen’s own base, 1.400 – see note above). Shaded band = noise floor. [Interactive version](#) →

Factor group	Effect range	Verdict
Generator model (M1)	−0.40 to −1.07 vs. sol	Dominant, real – ~10× any other factor
Adjacent-page fetch off (S5)	−0.067 (sol) / −0.133 (qwen)	Real for qwen, borderline for sol – keep on
Generic agent_harness factors (S6, S9–S12)	−0.134 to 0.000	Only 2 of 6 scored moves exceed the 1-topic noise floor (preview, wider-engines), both negative – none help
Brief-analyst model swap (M2)	−0.067 (all three, identical)	Noise-level, no real effect
Closure critic (S4 extension)	0.000	No effect

Table 4: Factor-effect summary by family. Noise floor: ± 0.067 (1 topic-grade of 15, $n = 1$ rerun).

4.3 Rubric axis breakdown

The official rubric groups its ~31 criteria into six axes: Communication Quality, Explicit Criteria, Implicit Criteria, Instruction Following, References & Citation Quality, and Synthesis of Information. Figure 3 breaks the standalone score down by axis for a representative subset of cells. **References & Citation Quality is the weakest axis in every single cell scored this session** (mean 0.196/2 across all 21 cells, range 0.00–0.67) – independent of generator model, structural configuration, or harness toggle. The number of criterion instances contributing to this axis varies

by cell and topic (the original validation run against brief-revise-iter1-exp15 pooled $n = 9$ instances across 15 topics for this axis specifically); no factor tested this session meaningfully moves it.

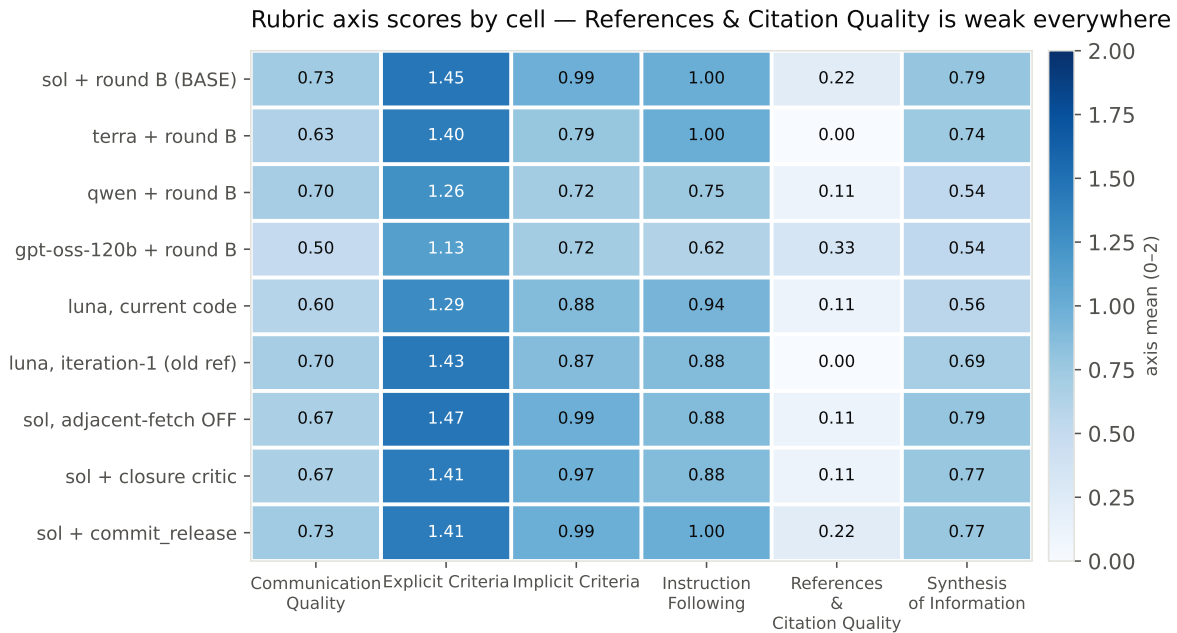


Figure 3: Rubric axis means (0–2) by cell, all 6 axes. References & Citation Quality stays pale throughout. [Interactive version →](#)

4.4 Arena confirmation

Figure 4 shows the per-topic outcome of the base cell against `aus_agent_v2` on the same 15 topics, both battle orders. Of 15 topics, 11 are order-consistent (“clean”: both orders agree, $\text{order_consistency} = \frac{11}{15} = 0.733$) – 4 clean wins, 7 clean losses – and 4 are not order-consistent (ambiguous: the judge’s verdict flips or ties depending on presentation order). The pooled win rate across all 30 individual battle judgments (both orders, not just the clean subset) is `aus_agent_v2` 60% / `brief_revise_agent` 40% (18–12). The base cell’s standalone-rubric lead over every other tested configuration does not translate into an arena win rate above 50% by either the clean or the pooled reading.

Arena, base cell vs `aus_agent_v2` — 15 topics, clean per-topic agreement

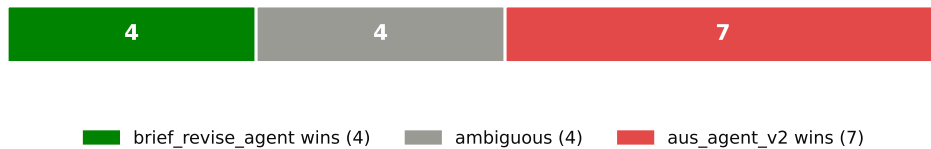


Figure 4: Arena outcome, base cell vs. `aus_agent_v2`, 15 topics, both battle orders. [Interactive version →](#)

5 Discussion

5.1 Why the hill-climb plateaued

Figure 2 makes the mechanism visible: once the generator model is already optimized (sol), the remaining structural and harness-level levers tested (Table 3) have an order of magnitude less headroom to move the standalone score. Ten of eleven post-model-selection moves landed at or below the 0.067 (one-topic) noise floor – not because they

were poorly chosen, but because the model-choice effect had already consumed most of the available variance at this topic-set size and rubric. This is a claim about the **tested** factor set (§4, Table 3): S1, S4-as-a-toggle, S7, S8, S13, and M3 were never varied this session (§3), so “local optimum” should be read as scoped to the eight factors actually scored, not the full 16-factor taxonomy.

5.2 Standalone score is not a full proxy for the arena objective

The base cell is the best standalone-rubric cell found and beats the fair (round-B-inclusive) luna comparison cell by 0.4 points (2.267 vs. 1.867, br-luna-current-code-exp15), yet loses the majority of arena battles against aus_agent_v2 (§5.4). Whatever makes aus_agent_v2 win head-to-head – plausibly completeness, coverage breadth, or claim density – is not fully captured by the official rubric’s per-criterion grading. A parallel, unresolved finding from this session: S5 (adjacent-page fetch) helped luna’s arena result (round B beat no-round-B 3W/9L/3A vs. 3W/10L/2A in the earlier rounds-B/C/D thread) but **hurt** its standalone score (1.867 vs. the pre-round-B reference 2.067), while helping both metrics for sol and qwen. This is a real disagreement between the two evaluation modes on at least one factor, for one model specifically, not an artifact to be explained away.

5.3 Model-choice replication is directionally solid, not a fixed number

Sol beats luna on a completely independent 15-topic set (2.133 vs. 1.933, +0.2), the same direction as the original comparison (+0.4 on the tuning topics), while the same-config, same-topic rerun of sol alone moved by 0.067 purely from stochasticity. The honest summary is that sol beats luna by **roughly 0.2 to 0.4 points** depending on topic sample – the direction replicates, the magnitude does not pin down to a single number at this sample size.

5.4 A separate, untouched weakness

References & Citation Quality sits near the bottom of every cell’s axis profile regardless of what factor was varied. This reads as a systemic issue in how brief_revise_agent constructs or formats citations and reference lists – a different investigation (citation format, reference- list construction logic) than anything this session’s factor sweep targeted, and plausibly higher-value than further generator or harness tuning given how flat the rest of the tested factor space turned out to be.

6 Recommendation

Use the base cell (gpt-5.6-sol, round-B structure, $k = 10$, luna brief-analyst/reviewer) as brief_revise_agent’s standing configuration – it is the best cell found among the 21 scored, and the eleven-move search around it on the tested factor set (Table 3) is exhausted. It is not yet competitive with aus_agent_v2 head-to-head. Untested factors that remain open questions, not ruled out: S7 (search_k, held at 10 throughout), M3 (reviewer model, never varied), S2/S3/S13 (schema/word-budget/screener variants, each tested in a different thread against a different baseline, not against the current sol base), and S8 (search_result_filter, not wired – would need a requirement-carrying search-tool schema first). Closing the arena gap most likely needs either (a) a structural change genuinely distinct from anything tried this session (e.g. a multi-candidate/ensemble mechanism, explicitly out of scope here per the brief to diverge from aus_agent_v2’s own architecture rather than re-derive it), or (b) treating brief_revise_agent as the lighter, cheaper alternative and reserving aus_agent_v2 for submissions where its cost is affordable.

7 Budget

The judging/generation budget cap was raised mid-session from an original \$50+\$50 split to \$400 once \$5.1’s model-effect result made further hill-climbing look worthwhile (§Method, Constraint discovered mid-session).

Line item	Cost	Basis
OpenAI generation (terra/sol/luna, ≈ 55.6 M tokens)	\$278	estimate – placeholder \$5/1M rate, no real metered rate card for gpt-5.6-*
Bedrock (gpt-oss-120b + qwen)	\$2.76	real – metered rate-card files
Sol design/thinking calls (7 calls)	\$1.01	real – exact printed API cost
Standalone + arena judging (≈ 390 calls)	\$58	estimate
Total	$\approx \\$340$	of the \$400 cap; $\approx \$60$ unspent

Table 5: Final cost breakdown. The OpenAI generation line is the single largest source of estimate uncertainty in this total.

8 Data and code

Interactive figures (hover tooltips, self-contained HTML, open directly in a browser, no server or network needed): `interactive/*.html`, alongside this PDF, generated by `make_interactive.py`. Every figure above links to its interactive counterpart.

All generated answers: `data/outputs/brief_revise_agent/` (by `run_id`). Standalone scores: `evaluation-results/factorial/<run_id>/` (each `scores.jsonl/summary.json`, per-topic grades truncated at 16,000 input characters, §Method). Arena judgments: `evaluation-results/factorial/arena-sol-vs-aus-agent-v2-exp15/`.

Sub-system manifests (named, reusable JSON configs under `src/systems/brief_revise_agent/subsystems/*.json`): only 5 of the 21 scored cells were materialized this way – the original 4 Block-1 model cells plus the 1 divergent-anchor cell (§5 of the narrative worklog). The 11 hill-climb moves in Table 3 and the 3 replication cells exist only as `run_ids` in the execution index below, not as named, reusable manifests.

Execution index (sub-system name $\rightarrow$ `run_id`, append-only): `evaluation-results/factorial/executions.jsonl`. Full narrative log with every intermediate decision, including the running move-count discrepancies reconciled in Table 3: `worklogs/2026-08-07-brief-revise-agent-llm-factorial-design.md`. Full factor taxonomy (source for §3): `worklogs/assets/2026-08-07-terra-factor-taxonomy-final.md`. New code this session: `commit_release_tool.py`, `REVIEW_PROMPT_WITH_CLOSURE` + `closure_check` (`prompts.py/review.py/agent.py/run.py`), five generic-factor CLI flags – all on branch `explore/new-agent-framework-system`.